

Rahul Sharma

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Education

George Mason University - College of Engineering & Computing Aug 2022 – Dec 2026
Bachelor of Science in Computer Science, GPA: 3.75 Fairfax, VA

- **Relevant Coursework:** Data Structures & Algorithms, Database Systems, Machine Learning, Software Engineering, Operating Systems

Experience

Undergraduate Teaching Assistant May 2026 – Present
George Mason University - Department of Computer Science Fairfax, VA

- CS 310 (Data Structures), Summer 2026: supported 100+ students across two sections through weekly office hours and Piazza, walking through Java lists, trees, hash tables, and graphs and debugging code that would not run.
- CS 405 (Ethics and Law in Computing), Fall 2026: grade analysis essays and the assignments building toward the semester-long mock trial, and score the in-class trial.

Founding Engineer Jun 2025 – Aug 2025
TheCollegeTech - Educational Platform Solutions

- Built a Learning Management System with a TypeScript/React frontend and a Django REST backend, including authentication, account management, and role-based access control on Supabase.
- Designed the course creation, enrollment, and content workflows, with paginated and filtered SQL queries.
- Set up CI/CD with automated tests, wrote the API documentation, and ran code reviews.

College of Engineering IT Technician Jun 2023 – Present
George Mason University - College of Engineering & Computing Fairfax, VA

- Maintain 10–15 computer labs (~30 machines each) at ~95% uptime through automated imaging, troubleshooting, and software optimization.
- Write Python, Bash, and PowerShell automation scripts for deployment and configuration across Windows, macOS, and Linux lab machines.
- Provide first-level support to faculty, staff, and graduate assistants, and author the documentation and ticketing-system content that reduced repeat tickets.

Projects

Drive | Go, React, TypeScript, PostgreSQL, S3 Multipart, Docker, Playwright Jul 2026 – Sep 2026

- Built a self-hosted file storage platform where a single Go binary serves the REST API and React/TypeScript app; file bytes go browser-to-storage over presigned S3 multipart URLs, never through the server.
- Designed a resumable upload protocol recording each confirmed part in PostgreSQL, so transfers survive dropped connections, closed tabs, and server crashes, resuming from the last confirmed part.
- Verified integrity with per-part MD5/ETag checks and ledger reconciliation before publish; measured 2 GiB uploads at ~77 MiB/s and an 11 GiB upload across 1,127 parts, with SHA-256-verified round trips.
- Deployed to production with Google OAuth and Argon2id authentication, per-IP rate limits, storage quotas, hourly garbage collection, and revocable share links with optional password, expiry, and a download cap counted once per recipient session; share tokens are stored only as SHA-256 hashes.

XPen\$e | Next.js, Supabase, NFC, Auth0, Google Maps API, Perplexity AI API Dec 2024

- Engineered a smart wallet app to maximize credit card rewards by detecting merchant data via NFC and EMV.
- Integrated Google Maps API for merchant location retrieval and Perplexity AI API for personalized reward recommendations.
- Implemented secure authentication with Auth0 and scalable user data storage with Supabase.

Technical Skills

Programming Languages: Java, Python, Go, C/C++, Kotlin, JavaScript, TypeScript
Web & App Development: React, Next.js, Django, Flask, FastAPI, Bootstrap, RESTful APIs
Frameworks & Libraries: Node.js, NumPy, JUnit
Databases: SQL (MySQL, PostgreSQL), MongoDB, Redis, Firebase, Supabase
Cloud & DevOps: Docker, AWS (EC2, S3), S3-compatible object storage, Microsoft Azure, Git, Linux